Selberg's Formula as a Riccati Flow, with an Application to a Sparse Mertens Certificate

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October 2026

Version 0.1.0 (release candidate)

Abstract

Wirsing showed in 1962–64, by elementary methods built on Selberg's symmetry formula, that $\psi(x) = x + O(x/\log x)$; Bombieri and Wirsing reached $O(x/(\log x)^A)$ for every A. We give a different proof of the mean-square form of the first bound. Let $f(t) = e^{-t}\psi(e^t) - 1$, and let $b(s)$ be the Laplace transform of the prime-power measure. On each horizontal line $s = \sigma + i\xi$, Selberg's formula becomes a Riccati equation for $\sigma b(\sigma + i\xi)$ as σ decreases. Positivity at the doubled frequency 2ξ keeps the solution away from the unstable fixed point -1. A Möbius substitution then turns it into a linear equation whose forcing term depends on the solution but stays small, which gives a resolvent bound uniform in ξ. Combined with a local density bound for primes from a quadratic sieve, this proves the energy bound $\int_0^T t^2 f(t)^2 \, \mathrm{d}t = O(T)$. The proof stays in the half-plane of absolute convergence and uses no analytic continuation.

The argument applies to the general coupled-source setting of Theorem 2.1: a positive measure ν with $\nu([0,t]) = t + O(1)$, coupled to a bounded source f by a first-order equation and a Selberg-type relation with bounded error. With a local density hypothesis it gives the energy bound; without it, the same method proves $f \in L^2(0,\infty)$. This supplies the hypotheses of Wirsing's abstract Tauberian theorem that the Selberg relation does not give directly. That theorem, which Wirsing stated without a detailed proof, then gives $f(t) = O(1/t)$, with a constant that may depend on the measure. As an application, the energy bound closes a sparse divisor–cofactor certificate for the Mertens function and gives $M(N) = O(N/\log N)$. These rates are classical. The contribution is the method.

Status and AI assistance. The proofs in this paper were written with the assistance of AI models and were checked by separate fresh-context AI proof audits, all using models from one AI provider. They have not been reviewed by a human expert or formally verified. Novelty is qualified in Section 16.

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1 Introduction

Let $\mu(n)$ be the Möbius function, $\Lambda(n)$ the von Mangoldt function, and

$$M(x) = \sum_{n \le x} \mu(n), \qquad \psi(x) = \sum_{n \le x} \Lambda(n), \qquad \vartheta(x) = \sum_{p \le x} \log p.$$

All logarithms are natural. A letter p or q used as an arithmetic index denotes a prime. We reserve ν for positive measures, so that the measure in the analytic argument is not confused with $\mu(n)$. Dirichlet convolution is denoted by $*_D$; convolution on the additive half-line or a specified circle is denoted by $*$. Constants implicit in O, $\ll$, and $\asymp$ are independent of the scale tending to infinity. They depend only on the stated hypothesis bounds, except in Corollary 8.3, where they may also depend on the particular pair (f, ν).

The main analytic result is Theorem 2.1. In the arithmetic case it applies to

$$f(t) = e^{-t}\psi(e^t) - 1, \qquad \nu = \sum_{n \ge 2} \frac{\Lambda(n)}{n} \delta_{\log n}, \qquad t \ge 0, \tag{1.1}$$

and gives

$$E(T) := \int_0^T t^2 |f(t)|^2 \, \mathrm{d}t \ll T \qquad (T \ge 1). \tag{1.2}$$

Equivalently, $\int_T^{2T} |f(t)|^2 \, \mathrm{d}t \ll T^{-1}$. This is a mean-square bound. The pointwise bound $f(t) = O(1/t)$, which is stronger, is due to Wirsing [12, 13].

How the proof works. Let $b(s)$ be the Laplace transform of ν. On a horizontal line $s = \sigma + i\xi$, Selberg's formula becomes the Riccati equation $\beta' = -\beta - \beta^2 + e$ for $\beta = \sigma b(\sigma + i\xi)$, where the derivative is taken as σ decreases and the error e is small (Section 3). The point $\beta = -1$ is an unstable fixed point. If β came close to it, positivity of ν would force $\operatorname{Re}\beta(\sigma, 2\xi)$ to stay near 1. The same Riccati equation at frequency 2ξ drives it down too fast for that to last. So β stays away from -1, uniformly in ξ (Lemma 3.1). The substitution $W = \beta/(1+\beta)$ cancels the quadratic term exactly, leaving a linear equation for W whose forcing depends on β but is small. Solving it bounds $\int |b(u+i\xi)|\,\mathrm{d}u$ along each line, which gives a resolvent for H, the Laplace transform of f (Section 4). The remaining work is to control H on a starting line and at high frequencies (Sections 5–7). This is where the density hypothesis (2.4) is used. For primes it follows from a quadratic sieve (Section 10).

Without the density hypothesis. Section 8 drops (2.4). The same argument then proves $f \in L^2(0,\infty)$. For a general positive measure, this supplies the hypotheses of Wirsing's abstract theorem (his Satz 2) that the Selberg relation does not give directly. That theorem then gives $f(t) = O(1/t)$ and (1.2), with constants that may depend on the pair (f, ν). This step relies on Wirsing's theorem, which he stated without a detailed proof. Nothing else in the paper depends on it.

The application. Sections 11–13 turn the energy bound into an exact finite identity with prime and semiprime coefficients. They handle discrete sampling, coprimality, and the finite cutoff explicitly. The results are

$$|M(N)| \ll \frac{N}{\log N}, \qquad \psi(x) = x + O\!\left(\frac{x}{(\log x)^{1/3}}\right). \tag{1.3}$$

These rates are classical.

What we think is new. The bounds are not new; the way they are proved is. We see three new ingredients.

1. *Selberg's formula as a differential equation in σ.* The formula is solved as a quantitative Riccati equation for the Laplace transform, inside the half-plane of absolute convergence. The elementary remainder proofs we examined (Kuhn, Wirsing, Bombieri) work with real-variable convolution inequalities; Bombieri uses the Laplace transform only to prove a convolution identity on the real axis. The analytic proofs we examined (Wiener–Geller, the 3-4-1 argument, Tao) use the transform on the boundary line and give no rate (Section 16.3). Our argument uses the transform but never reaches the boundary, and it gives a rate.

2. *Positivity at 2ξ as a barrier for the flow.* The inequality is the one behind the 3-4-1 argument. What is new is how it is used. It keeps the flow a fixed distance from -1 for a fixed time, uniformly in ξ, and this replaces analyticity.

3. *The linearized resolvent and what it gives.* The substitution $W = \beta/(1+\beta)$ removes the quadratic term of the flow; the remaining forcing depends on the solution but is small, and the resulting resolvent is bounded uniformly in ξ. In the general coupled-source setting it gives $f \in L^2$ with no density hypothesis. Through Wirsing's abstract theorem, this makes his pointwise bound $f(t) = O(1/t)$ available beyond the arithmetic case (Section 8).

We did not find any of these in the sources compared in Section 16.3. That comparison was not exhaustive, so priority is not settled.

What the proof uses. Laplace transforms in their half-plane of absolute convergence, Plancherel's theorem, and the Poisson kernel. It uses no analytic continuation and no zero-free region, no Hahn–Banach theorem, and no imported bound for the error in the prime number theorem. All arithmetic inputs are proved or cited explicitly.

What is not proved. The resolvent bound holds for a source built from the same measure as its convolution multiplier. It is not a bound for arbitrary vectors. Sections 14 and 15 record the finite eigenstructure, singular-vector formulation, and recursions that motivated the work, and state which assertions remain open. A related estimate on the circle, which does not by itself close the argument, is in Appendix A.

2 The abstract weighted energy theorem

Theorem 2.1 (Weighted energy for a coupled positive source)**.** *Let ν be a positive, locally finite measure on $[0,\infty)$ with no atom at zero, and set $B(t) = \nu([0,t])$. Suppose f is real, bounded, right-continuous, and locally of bounded variation, with $f(0) = -1$. Assume*

$$B(t) = t + O(1), \tag{2.1}$$

$$\nu = df + (1+f)\,\mathrm{d}t \quad \text{on } (0,\infty), \tag{2.2}$$

$$R(t) := tf(t) + \int_{[0,t]} f(t-u)\nu(\mathrm{d}u) = O(1). \tag{2.3}$$

Assume further that, for $t \geq 0$ and $0 < h \leq 1$,

$$\nu((t,t+h]) \leq C_\ell h + q(t), \qquad 0 \leq q(t) \leq C_q(1+t)^2 e^{-t/2}. \tag{2.4}$$

Then $E(T) \ll T$ for all $T \geq 1$, with a constant depending only on the hypothesis bounds.

The derivative in (2.2) is the Stieltjes derivative on the positive half-line. When a function is extended by zero to negative arguments, its boundary value at zero is accounted for separately. In particular, integrating (2.2) gives the bounded primitive

$$F(t) := \int_0^t f(u)\,\mathrm{d}u = B(t) - t - f(t) - 1 = O(1). \tag{2.5}$$

Right-continuity fixes representatives at atoms; all subsequent L^2 statements are independent of changes on null sets.

For $s = \sigma + i\xi$, $\sigma > 0$, define

$$b(s) = \int_{[0,\infty)} e^{-st}\nu(\mathrm{d}t), \quad H(s) = \int_0^\infty e^{-st} f(t)\,\mathrm{d}t, \quad P(s) = \int_0^\infty e^{-st} R(t)\,\mathrm{d}t. \tag{2.6}$$

These integrals and their derivatives converge absolutely where used. Our Fourier transform is $\widehat{u}(\xi) = \int_{\mathbb{R}} e^{-i\xi t} u(t)\,\mathrm{d}t$, so $\|\widehat{u}\|_2^2 = 2\pi\,\|u\|_2^2$.

3 The scale equation and two-frequency positivity

3.1 The exact Riccati equation

Integration by parts, with the boundary value $f(0) = -1$, gives

$$b(s) = (s+1)\left(H(s) + \frac{1}{s}\right), \qquad P(s) = -H'(s) + b(s)H(s). \tag{3.1}$$

For example, $\int_{(0,\infty)} e^{-st} df(t) = 1 + sH(s)$; the constant 1 must not be dropped. Eliminating H gives

$$-b'(s) + b(s)^2 = A(s)b(s) + D(s) + (s+1)P(s), \tag{3.2}$$

where

$$A(s) = 1 + \frac{1}{s(s+1)}, \qquad D(s) = \frac{s+1}{s^2}.$$

Fix $\xi \neq 0$ and $a > 0$, write $\sigma = ae^{-v}$, and put $\beta(\sigma, \xi) = \sigma b(\sigma + i\xi)$. Equation (3.2) becomes

$$\frac{\mathrm{d}\beta}{\mathrm{d}v} = -\beta - \beta^2 + e(\sigma, \xi), \qquad e = \sigma A(s)\beta + \sigma^2 D(s) + \sigma^2(s+1)P(s). \tag{3.3}$$

Let $g(\xi) = 1 + |\xi| + |\xi|^{-1}$. There is a constant $C_e \geq 1$ such that

$$|e(\sigma, \xi)| \leq C_e \sigma g(\xi), \qquad 0 < \sigma \leq \min(1, |\xi|). \tag{3.4}$$

Indeed $|P(s)| \leq \|R\|_\infty / \sigma$, $|A(s)| \leq 1 + |\xi|^{-1}$, and $|D(s)| \leq (2 + |\xi|)/|\xi|^2$. Positivity and (2.1) give

$$|\beta(\sigma, \xi)| \leq c_\sigma := \sigma b(\sigma), \qquad |c_\sigma - 1| \leq C_B \sigma, \tag{3.5}$$

where $C_B \geq 1$ bounds $|B(t) - t|$. To see the last estimate, use $c_\sigma = \sigma^2 \int_0^\infty e^{-\sigma t} B(t)\, \mathrm{d}t$. The term $\sigma^2/|\xi|^2$ in the defect is at most $\sigma/|\xi|$ in the stated range. The high-frequency loss in (3.4) is retained.

3.2 Uniform exclusion of the multiplier -1

Fix, once and for all,

$$0 < c \leq \min\left\{10^{-3}, \frac{1}{1000 C_B}, \frac{1}{2000 C_e}\right\}, \qquad a(\xi) = \frac{c}{g(\xi)}. \tag{3.6}$$

For $0 < \sigma \leq a(\xi)$, both ξ and 2ξ lie in the range of (3.4), and

$$|c_\sigma - 1| \leq 10^{-3}, \qquad |e(\sigma, \xi)| \leq 10^{-3}, \qquad |e(\sigma, 2\xi)| \leq 10^{-3}. \tag{3.7}$$

Here $g(2\xi) \leq 2g(\xi)$ supplies the second-frequency bound. No condition $\sigma \leq a(2\xi)$ is needed.

The measure $e^{-\sigma t}\nu(\mathrm{d}t)/b(\sigma)$ is a probability measure. For $U = e^{-i\xi t}$, the identity $\operatorname{Re} U^2 = 2(\operatorname{Re} U)^2 - 1$ and the nonnegativity of variance give

$$\operatorname{Re}\beta(\sigma, 2\xi) \geq \frac{2(\operatorname{Re}\beta(\sigma, \xi))^2}{c_\sigma} - c_\sigma. \tag{3.8}$$

Lemma 3.1. *For every $\xi \neq 0$ and $0 < \sigma \leq a(\xi)$, $|1 + \beta(\sigma, \xi)| \geq 1/100$.*

Proof. Suppose the distance is less than $1/100$ at a starting value σ_*. Follow (3.3) for $\sigma = \sigma_* e^{-v}$, $0 \le v \le \log 2$. For $y = 1 + \beta(\sigma, \xi)$, $y' = -\beta y + e$. Since $|\beta| \le 1.001 < 2$, Gronwall's inequality yields

$$|y(v)| \le e^{2v}|y(0)| + \frac{e^{2v} - 1}{2000} < \frac{4}{100} + \frac{3}{2000} < \frac{1}{20}.$$

Hence $\operatorname{Re}\beta(\sigma, \xi) < -19/20$. Equation (3.8) gives $u(v) := \operatorname{Re}\beta(\sigma, 2\xi) \ge 4/5$ throughout the interval. The same scale equation at 2ξ and (3.7) imply

$$u' \le -u - 2u^2 + |\beta(\sigma, 2\xi)|^2 + 10^{-3} \le -\frac{4}{5} - 2\left(\frac{4}{5}\right)^2 + (1.001)^2 + 10^{-3} < -1.$$

This is impossible: u must remain in $[0.8, 1.001]$, an interval of length 0.201, for time $\log 2 > 0.69$. □

4 A uniform resolvent along the actual source

For a fixed nonzero ξ, set $a = a(\xi)$, $\sigma = ae^{-v}$, and $W = \beta/(1 + \beta)$. The exact cancellation of the quadratic term gives

$$W' + W = \frac{e}{(1 + \beta)^2}. \tag{4.1}$$

Lemma 3.1 bounds the denominator away from zero. Also $|W(0)| \le 101$ and $|e| \le C_e c e^{-v}$, so

$$|W(v)| \ll (1 + v)e^{-v},$$
$$|\beta(\sigma, \xi)| \ll \frac{\sigma}{a}\left(1 + \log\frac{a}{\sigma}\right), \tag{4.2}$$
$$|b(\sigma + i\xi)| \ll \frac{1}{a}\left(1 + \log\frac{a}{\sigma}\right). \tag{4.3}$$

For the second line use $\beta = (1 + \beta)W$ and $|1 + \beta| \le 2.001$. In particular,

$$\int_0^{a(\xi)} |b(u + i\xi)|\,\mathrm{d}u \ll 1 \tag{4.4}$$

uniformly in $\xi \ne 0$.

Solving the actual source equation $H' = bH - P$ along the horizontal line gives

$$\begin{aligned} H(\sigma + i\xi) = {} & \exp\left(-\int_\sigma^a b(v + i\xi)\,\mathrm{d}v\right) H(a + i\xi) \\ & + \int_\sigma^a \exp\left(-\int_\sigma^u b(v + i\xi)\,\mathrm{d}v\right) P(u + i\xi)\,\mathrm{d}u. \end{aligned} \tag{4.5}$$

Differentiation in σ verifies both the sign and integration direction. Writing $H_*(\xi) = H(a(\xi) + i\xi)$, (4.4) implies

$$|H(\sigma + i\xi)| \ll |H_*(\xi)| + \int_\sigma^{a(\xi)} |P(u + i\xi)|\,\mathrm{d}u. \tag{4.6}$$

The initial-source term H_* is retained. Its control is an essential part of the proof, not an independently assumed smallness condition.

5 Translation estimates and weighted Fourier tails

Lemma 5.1. *For $k \in \{0,1\}$, put $U_{\sigma,k}(t) = t^k e^{-\sigma t} f(t)$ on $[0,\infty)$ and zero on $(-\infty,0)$. For $0 < \sigma \le 1$ and $0 < h \le 1$,*

$$\|U_{\sigma,k}(\cdot - h) - U_{\sigma,k}\|_2^2 \ll h^2 \sigma^{-2k-1} + h. \tag{5.1}$$

Proof. Solving (2.2) over $(t, t+h]$ gives

$$f(t+h) - f(t) = (e^{-h} - 1)(1 + f(t)) + \int_{(t,t+h]} e^{u-t-h} \nu(\mathrm{d}u).$$

Consequently $|f(t+h) - f(t)|$ is bounded both by $Ch + \nu((t,t+h])$ and by $Ch + q(t)$. Multiply these two bounds. With $w(t) = (1+t)^{2k} e^{-2\sigma t}$, the resulting four terms are controlled by

$$\begin{aligned} h^2 \int_0^\infty w(t)\,\mathrm{d}t &\ll h^2 \sigma^{-2k-1}, \\ h \int_0^\infty w(t)\nu((t,t+h])\,\mathrm{d}t &\ll h^2 \sigma^{-2k-1}, \\ h \int_0^\infty w(t) q(t)\,\mathrm{d}t &\ll h, \\ \int_0^\infty w(t) q(t) \nu((t,t+h])\,\mathrm{d}t &\ll h. \end{aligned}$$

For the second line, Tonelli's theorem restricts t to an interval of length at most h adjacent to the integration variable u. On that interval $w(t) \ll (1+u)^{2k} e^{-2\sigma u}$. The measure moment is $O(\sigma^{-2k-1})$ by $B(u) = u + O(1)$. For the fourth line use the majorant $(1+u)^{2k+2} e^{-u/2}$ for $w(t)q(t)$ on the same interval; its integral against ν is finite uniformly in σ. This uses the majorant of q, not any unasserted regularity of q itself.

For the change of weight $v_k(t) = t^k e^{-\sigma t}$, Cauchy–Schwarz gives $\int_0^\infty |v_k(t+h) - v_k(t)|^2\,\mathrm{d}t \le h^2 \|v_k'\|_2^2$. The latter is $O(h^2\sigma)$ for $k = 0$ and $O(h^2/\sigma)$ for $k = 1$, both within the stated bound. Finally the boundary strip at zero costs $O(h^{2k+1})$. These estimates prove (5.1). □

Average the Fourier translation identity over $0 < h < 2/K$, $K \ge 2$. For $|\xi| \ge K$ the averaged multiplier is

$$2\left(1 - \frac{\sin(2\xi/K)}{2\xi/K}\right) \ge 1.$$

Since the transforms of $U_{\sigma,0}$ and $U_{\sigma,1}$ are respectively $H(\sigma + i\xi)$ and $-H'(\sigma + i\xi)$, we obtain

$$\int_{|\xi|\ge K} |H(\sigma + i\xi)|^2\,\mathrm{d}\xi \ll \frac{1}{\sigma K^2} + \frac{1}{K}, \tag{5.2}$$

$$\int_{|\xi|\ge K} |H'(\sigma + i\xi)|^2\,\mathrm{d}\xi \ll \frac{1}{\sigma^3 K^2} + \frac{1}{K}. \tag{5.3}$$

The terms $1/K$ explicitly retain the atomic and initial-boundary costs.

6 The variable starting line

Lemma 6.1. *The initial-source term satisfies*

$$\int_{r\le|\xi|\le 2r} |H_*(\xi)|^2\,\mathrm{d}\xi \ll r \qquad (0<r\le 1), \tag{6.1}$$

$$\int_{Q\le|\xi|\le 2Q} |H_*(\xi)|^2\,\mathrm{d}\xi \ll Q^{-1} \qquad (Q\ge 1). \tag{6.2}$$

Proof. By (2.5), $H(s)=s\int_0^\infty e^{-st}F(t)\,\mathrm{d}t$, whence $|H(s)|\ll |s|/\operatorname{Re} s$. On a low shell, $a(\xi)\asymp cr$ and $|a(\xi)+i\xi|\asymp r$. This proves (6.1).

For a positive high shell $Q\le\xi\le 2Q$, $Q\ge 1$, we have

$$\frac{c}{4Q}\le a(\xi)\le\frac{c}{Q}.$$

Set $\tau=c/(8Q)$ and $d(\xi)=a(\xi)-\tau$. Then $d_1:=c/(8Q)\le d(\xi)\le 7c/(8Q)=:d_2$. For the Poisson kernel $\mathcal{P}_d(y)=d/[\pi(d^2+y^2)]$,

$$H(a(\xi)+i\xi)=\mathcal{P}_{d(\xi)}*[H(\tau+i\cdot)](\xi). \tag{6.3}$$

For fixed d this follows by Fourier transformation: convolution by $\mathcal{P}_d$ multiplies the original half-line function by $e^{-d|t|}=e^{-dt}$. Both sides are continuous on the positive half-plane, so they can be evaluated at each indicated $(d(\xi),\xi)$.

Jensen's inequality bounds the square in (6.3) by the Poisson average of $|H(\tau+i\cdot)|^2$. The dependence of d on ξ does not invalidate integration, because

$$\mathcal{P}_{d(\xi)}(y)\le\frac{d_2}{d_1}\mathcal{P}_{d_1}(y)\le 7\mathcal{P}_{d_1}(y). \tag{6.4}$$

Thus frequencies $\eta\in[Q/2,3Q]$ contribute at most

$$C\int_{|\eta|\ge Q/2}|H(\tau+i\eta)|^2\,\mathrm{d}\eta\ll Q^{-1}$$

by (5.2) when $Q\ge 4$. Outside this enlarged shell, $|\xi-\eta|\ge Q/2$, so the kernel is $O(Q^{-3})$ and its integral over $\xi\in[Q,2Q]$ is $O(Q^{-2})$. That contribution is at most

$$\frac{C}{Q^2}\int_{\mathbb{R}}|H(\tau+i\eta)|^2\,\mathrm{d}\eta\ll\frac{1}{Q^2\tau}\ll Q^{-1},$$

using bounded f and Plancherel. The negative shell is identical. The bounded range $1\le Q<4$ follows from $|H(a+i\xi)|\ll 1/a$ with adjusted constants. This proves (6.2), including off-shell leakage. □

7 Closure of the weighted estimate

We now prove Theorem 2.1. Fix

$$0<\sigma\le\sigma_0:=c/16,\qquad \eta_0=4\sigma/c,\qquad K=c/(4\sigma). \tag{7.1}$$

Then $\eta_0 \le 1/4$ and $K \ge 4$. Throughout $\eta_0 \le |\xi| \le K$ we have $\sigma \le a(\xi)$: use $g(\xi) \le 3/|\xi|$ for $|\xi| \le 1$ and $g(\xi) \le 3|\xi|$ for $|\xi| \ge 1$. We claim

$$\int_{\mathbb{R}} |H'(\sigma + i\xi)|^2 \, \mathrm{d}\xi \ll \sigma^{-1}. \tag{7.2}$$

For $|\xi| \le \eta_0$, differentiating the primitive formula gives $|H'(s)| \ll \sigma^{-1} + |s|\sigma^{-2} \ll \sigma^{-1}$. The interval has length $O(\sigma)$, so its squared integral is $O(\sigma^{-1})$. For $|\xi| \ge K$, (5.3) gives the same bound.

In the middle range, $H' = bH - P$ and

$$\int_{\mathbb{R}} |P(u + i\xi)|^2 \, \mathrm{d}\xi = 2\pi \int_0^\infty e^{-2ut} |R(t)|^2 \, \mathrm{d}t \ll u^{-1}. \tag{7.3}$$

It suffices to prove

$$\int_{\eta_0 \le |\xi| \le K} |\beta(\sigma, \xi) H(\sigma + i\xi)|^2 \, \mathrm{d}\xi \ll \sigma. \tag{7.4}$$

Apply (4.6), retaining both terms.

The initial-source contribution on a low dyadic shell $[r, 2r]$ is, by (4.2) and (6.1), at most

$$C \frac{\sigma^2}{r} \left(1 + \log \frac{r}{\sigma}\right)^2.$$

Summing upwards from $r = \eta_0$ costs $O(\sigma)$, since $\sum_{j \ge 0} 2^{-j}(1 + j)^2 < \infty$. On a high shell $[Q, 2Q]$, (6.2) gives at most

$$C\sigma^2 Q \left(1 + \log \frac{1}{\sigma Q}\right)^2.$$

Summing downwards from a shell comparable to K again costs $O(\sigma)$. End shells can be intersected with the middle range. All comparison constants depend only on the fixed c.

For the forcing contribution, abbreviate $a = a(\xi)$ and use Cauchy–Schwarz in u:

$$|\beta|^2 \left(\int_\sigma^a |P(u + i\xi)| \, \mathrm{d}u \right)^2 \ll \frac{\sigma^2}{a} \left(1 + \log \frac{a}{\sigma}\right)^2 \int_\sigma^a |P(u + i\xi)|^2 \, \mathrm{d}u. \tag{7.5}$$

For $\sigma \le u \le a$,

$$\frac{(1 + \log(a/\sigma))^2}{a} \le 2 \frac{(1 + \log(u/\sigma))^2}{u}. \tag{7.6}$$

Indeed $(1 + \log x)^2/x$ increases on $[1, e]$ from 1 to $4/e < 2$ and decreases thereafter. Tonelli's theorem, (7.3), and $a(\xi) \le c$ bound the integral of (7.5) over the middle range by

$$C\sigma^2 \int_\sigma^c \frac{(1 + \log(u/\sigma))^2}{u^2} \, \mathrm{d}u \le C\sigma \int_1^\infty \frac{(1 + \log y)^2}{y^2} \, \mathrm{d}y \ll \sigma.$$

This proves (7.4) and (7.2). The forcing was charged in L^2 across frequencies; replacing it everywhere by its pointwise bound C/u would discard this control.

Finally, Plancherel gives

$$\int_0^\infty t^2 e^{-2\sigma t} |f(t)|^2 \, \mathrm{d}t = \frac{1}{2\pi} \int_{\mathbb{R}} |H'(\sigma + i\xi)|^2 \, \mathrm{d}\xi \ll \sigma^{-1}.$$

For $T \ge \sigma_0^{-1}$, set $\sigma = 1/T$ and use $e^{-2t/T} \ge e^{-2}$ on $[0, T]$. For $1 \le T < \sigma_0^{-1}$, boundedness of f gives $E(T) \le \|f\|_\infty^2 \, T^3/3 \ll T$ with the permitted constant. This completes the abstract proof.

8 Removing the density hypothesis

The separate hypothesis (2.4) can be removed if Wirsing's abstract Tauberian theorem is used as an additional input. The connecting estimate is global square-integrability of the actual source. We prove it below from (2.1)–(2.3), using only the parts of the preceding resolvent argument that do not use (2.4). This gives a second route to the energy conclusion; the proof in Sections 3–7 continues to give that conclusion without invoking Wirsing.

Lemma 8.1 (An interval bound from positivity)**.** *Under the hypotheses of Theorem 2.1 other than* (2.4)*, for every fixed $A > 0$,*

$$\nu((t, t+h]) \le 2h + \frac{C_A}{t} \qquad (t \ge 2,\ 0 < h \le A). \tag{8.1}$$

In particular, $\nu((t, t+h]) \le 2h + C/(1+t)$ for $t \ge 0$ and $0 < h \le 1$.

Proof. The function R is locally of bounded variation. Differentiation on the positive half-line gives the measure identity

$$dR = (t-1)\nu + \nu * \nu + (f - t - B - R)\,\mathrm{d}t. \tag{8.2}$$

Indeed $d(f*\nu) = df*\nu - \nu$, with the last term supplied by $f(0) = -1$. Substitute $df = \nu - (1+f)\,\mathrm{d}t$ and $f*\nu = R - tf$ to obtain (8.2). Positivity, integrated over $(t, t+h]$, now gives

$$(t-1)\nu((t,t+h]) \le R(t+h) - R(t) + \int_t^{t+h} (u + B(u) + R(u) - f(u))\,\mathrm{d}u \le 2th + h^2 + Ch + 2\,\|R\|_\infty .$$

Division by $t-1$ proves (8.1); the bounded initial interval is absorbed by increasing the constant. □

Theorem 8.2 (Square-integrability without a separate density input)**.** *Assume all the base hypotheses of Theorem 2.1, including* (2.1)–(2.3)*, but omit* (2.4)*. Then*

$$\int_0^\infty |f(t)|^2\,\mathrm{d}t < \infty. \tag{8.3}$$

The integral is bounded in terms of the base hypothesis bounds.

Proof. Write $\ell_\sigma = \log(e/\sigma)$ for $0 < \sigma \le 1$. Use Lemma 8.1 in the proof of Lemma 5.1, with $k = 0$ and $q(t) = C/(1+t)$. The two bounds for the source increment are now $Ch + \nu((t, t+h])$ and $Ch + C/(1+t)$. Multiplication and Tonelli give

$$\|U_{\sigma,0}(\cdot - h) - U_{\sigma,0}\|_2^2 \ll h^2/\sigma + h\ell_\sigma \qquad (0 < h \le 1). \tag{8.4}$$

For the two terms containing q, use

$$\int_0^\infty \frac{e^{-2\sigma u}}{1+u}\big(\,\mathrm{d}u + \nu(\mathrm{d}u)\big) \ll \ell_\sigma.$$

The measure estimate follows by dividing into unit intervals and using $B(u) = u + O(1)$. Weight changes and the strip at zero cost $O(h^2\sigma + h)$, within (8.4). Averaging the Fourier translation identity as before yields

$$\int_{|\xi|\ge K} |H(\sigma + i\xi)|^2\,\mathrm{d}\xi \ll \frac{1}{\sigma K^2} + \frac{\ell_\sigma}{K}, \qquad K \ge 2. \tag{8.5}$$

The scale equation, two-frequency exclusion, and source resolvent in Sections 3–4 used only the base hypotheses. Retain their c, $a(\xi)$, and $H_*(\xi)$. The low-shell bound (6.1) also used only boundedness of F. On a high shell, repeat the Poisson argument of Lemma 6.1, now with (8.5) on the line $\tau = c/(8Q)$. The enlarged shell costs $O(\log(2Q)/Q)$, and the off-shell leakage is still $O(1/Q)$. The bounded ratio of widths in (6.4) is unchanged. Thus

$$\int_{Q\le|\xi|\le 2Q} |H_*(\xi)|^2 \,\mathrm{d}\xi \ll \frac{\log(2Q)}{Q} \qquad (Q \ge 1). \tag{8.6}$$

The tail calculation uses $Q \ge 4$; the remaining compact range $1 \le Q < 4$ follows from boundedness of f and a positive lower bound for $a(\xi)$ there. Both this high-shell series and the low-shell series $O(r)$ converge dyadically, so $\|H_*\|_2 \ll 1$.

Fix $0 < \sigma \le c/16$ and let $D_\sigma = \{4\sigma/c \le |\xi| \le c/(4\sigma)\}$. Here $\sigma \le a(\xi)$, so (4.6) applies. Its forcing term is bounded in L^2 by Minkowski's inequality:

$$\begin{aligned}\left\| \mathbf{1}_{D_\sigma} \int_\sigma^{a(\xi)} |P(u+i\xi)| \,\mathrm{d}u \right\|_2 &\le \int_\sigma^c \left\| \mathbf{1}_{D_\sigma} \mathbf{1}_{\{u\le a(\xi)\}} P(u+i\xi) \right\|_2 \mathrm{d}u \\ &\ll \int_\sigma^c u^{-1/2} \,\mathrm{d}u \ll 1. \end{aligned} \tag{8.7}$$

The estimate on P is precisely (7.3), which requires only bounded R. Hence $\int_{D_\sigma} |H(\sigma+i\xi)|^2 \,\mathrm{d}\xi \ll 1$. On $|\xi| < 4\sigma/c$, the bounded primitive gives $|H(s)| \ll |s|/\sigma \ll 1$, at squared-integral cost $O(\sigma)$. On $|\xi| > c/(4\sigma)$, (8.5) costs $O(\sigma\ell_\sigma)$. We have proved

$$\sup_{0<\sigma\le c/16} \int_{\mathbb{R}} |H(\sigma+i\xi)|^2 \,\mathrm{d}\xi < \infty.$$

Plancherel and monotone convergence as $\sigma \downarrow 0$ prove (8.3). This proof has not used the weighted energy theorem or any pointwise PNT consequence. □

Corollary 8.3 (A bootstrap using Wirsing)**.** *Under the hypotheses of Theorem 8.2, Wirsing's Satz 2 [13, Section 5, p. 15] gives*

$$f(t) = O(t^{-1}), \qquad E(T) = O(T). \tag{8.8}$$

In the arithmetic instance, this yields $\psi(x) = x + O(x/\log x)$. This corollary depends on Wirsing's Satz 2, which he stated without a detailed proof [13, p. 15]. The direct proof of the weaker pointwise rate in Corollary 13.3 does not depend on it.

Proof. First $f(t) \to 0$. The source equation and Lemma 8.1 give $|f(t+h) - f(t)| \le Ch + C/(1+t)$ for $0 < h \le 1$. If $|f(t_j)| \ge \varepsilon$ at arbitrarily large t_j, this inequality gives disjoint forward intervals of a fixed positive length on which $|f| \ge \varepsilon/2$, contradicting (8.3). This also treats the right-continuous values at atoms.

Put $g = -f$ and fix $\delta > 0$ with $\gamma := 1 + \delta < \sqrt{2}$. For all sufficiently large t, $|f(u)| \le \delta$ for $u \ge t$. For each fixed A the exact increment identity and (8.1) give

$$-(1+\delta)h \le f(t+h) - f(t) = \nu((t,t+h]) - h - \int_t^{t+h} f(u) \,\mathrm{d}u \le (1+\delta)h + C_A/t \qquad (0 < h \le A).$$

Taking $A = \max(1, 2\|f\|_\infty)$ handles all larger h by boundedness. This holds for $t \ge t_\delta$, where $t_\delta \ge 2$ depends on the pair. For $t < t_\delta$, $|g(t+h) - g(t)| \le 2\|f\|_\infty \le 2\|f\|_\infty t_\delta/t$. Consequently

$$|g(t+h) - g(t)| \le \gamma h + O(t^{-1}), \qquad h > 0. \tag{8.9}$$

Also $\int_{(0,T]} |dg| \le B(T) + \int_0^T |1+f|\,\mathrm{d}t = O(T)$. Finally, expanding $f * \nu$ using the source equation gives the exact identity

$$tg - g * dg = F + f * f - R = O(1), \tag{8.10}$$

because $|f * f| \le \|f\|_2^2$ and both F and R are bounded. Thus g has the total-variation bound, fluctuation bound with $\gamma < \sqrt{2}$, and self-convolution equation required by Satz 2, with Wirsing's exponent $\beta = 1$ (unrelated to the variable β of Section 3).

Wirsing uses a zero extension at $0-$ and allows the countably many points where a Stieltjes convolution is undefined. Adding the initial atom to our derivative convention changes $g * dg$ by $g(0)g(t)$, a bounded term. If J is the countable jump set of g, simultaneous interior jumps occur only for $t \in J + J$. Outside this countable set the Riemann–Stieltjes and measure convolutions agree; the printed exception covers the remaining points. The upper-endpoint value agrees because g is right-continuous at zero. Therefore his theorem applies and proves (8.8). The pointwise implied constant here is asserted for each fixed admissible pair; uniform dependence solely on the original hypothesis bounds is not claimed by this bootstrap. □

9 Classical arithmetic inputs and the coupled source

We record the classical arithmetic ingredients to make clear where the analytic hypotheses come from. None is a PNT remainder assumption.

Lemma 9.1 (Chebyshev and the harmonic prime-power sum)**.** *For $x \ge 2$,*

$$\vartheta(x) \ll x, \qquad \psi(x) \ll x, \qquad \sum_{n \le x} \frac{\Lambda(n)}{n} = \log x + O(1). \tag{9.1}$$

Also $\pi(x) \ll x/\log x$.

Proof. Every prime in $(n, 2n]$ divides $\binom{2n}{n}$, so $\vartheta(2n) - \vartheta(n) \le 2n \log 2$. Summing at powers of two and using monotonicity gives $\vartheta(x) \ll x$. The identity $\psi(x) = \sum_{a \ge 1} \vartheta(x^{1/a})$ then gives $\psi(x) \ll x$: the $a \ge 2$ terms are $O(\sqrt{x}\log x) = O(x)$.

From $\sum_{d|n} \Lambda(d) = \log n$,

$$\sum_{d \le x} \Lambda(d) \left\lfloor \frac{x}{d} \right\rfloor = \sum_{n \le x} \log n = x \log x - x + O(\log(2x)).$$

Replacing the floor by x/d costs at most $\psi(x) = O(x)$, which proves the harmonic estimate. Finally primes greater than $\sqrt{x}$ each contribute at least $(\log x)/2$ to $\vartheta(x)$; the remaining count is at most $\sqrt{x}$. This proves the bound on π. □

Lemma 9.2 (Selberg's symmetry formula)**.** *For $x \ge 2$,*

$$\sum_{n \le x} \big(\Lambda(n)\log n + (\Lambda *_D \Lambda)(n)\big) = 2x \log x + O(x). \tag{9.2}$$

Proof. Here is the elementary inversion argument, in the form discussed in [15, Section 1]. Put $\Lambda_2(n) = \sum_{d|n} \mu(d) \log^2(n/d)$. The identity $\Lambda_2 = \Lambda \log + \Lambda *_D \Lambda$ follows by applying logarithmic differentiation to $\mathbf{1} *_D \Lambda = \log$ and then convolving with μ.

Integral comparison and partial summation give constants c_1, c_2, c_3 with

$$\frac{1}{y}\sum_{n\le y}\log^2 n = \log^2 y - 2\log y + 2 + O\!\left(\frac{\log^2(2+y)}{y}\right),$$

$$\sum_{n\le y}\frac{1}{n} = \log y + c_1 + O(y^{-1}),$$

$$\sum_{n\le y}\frac{\log(y/n)}{n} = \tfrac{1}{2}\log^2 y + c_2\log y + c_3 + O\!\left(\frac{\log(2+y)}{y}\right).$$

For the last line one may integrate the preceding harmonic estimate; its $O(1/y)$ remainder has a convergent integral against dy/y. Consequently there are constants c_4, c_5 such that

$$\frac{1}{y}\sum_{n\le y}\log^2 n = \sum_{m\le y}\frac{2\log(y/m) + c_4 + c_5/(y/m)}{m} + O\!\left(\frac{\log^2(2+y)}{y}\right).$$

Insert $y = x/d$ and sum with coefficient $\mu(d)/d$. The error is bounded by $x^{-1}\sum_{d\le x}\log^2(2+x/d) = O(1)$, as follows by dyadic grouping. Grouping the main term by $k = dm$ and using $\sum_{d\mid k}\mu(d) = \mathbf{1}_{k=1}$ leaves $2\log x + c_4 + c_5/x$. Multiplication by x proves the lemma. □

Proposition 9.3. *The source and measure in* (1.1) *satisfy* (2.1)–(2.3).

Proof. The jump of f at $\log n$ is $\Lambda(n)/n$, and between jumps $f' = -(1+f)$. Also $f(0) = -1$ and ν has no atom at zero. Thus (2.2) holds with the stated endpoint convention. Lemma 9.1 supplies bounded f and $B(t) = t + O(1)$.

Put $x = e^t$. Partial summation and Chebyshev give

$$\sum_{n\le x}\Lambda(n)\log n = \psi(x)\log x - \int_1^x \frac{\psi(u)}{u}\,\mathrm{d}u = \psi(x)\log x + O(x).$$

Moreover

$$\frac{1}{x}\sum_{n\le x}(\Lambda *_D \Lambda)(n) = \sum_{k\le x}\frac{\Lambda(k)}{k}\big(1 + f(\log x - \log k)\big) = B(t) + (f*\nu)(t).$$

Divide (9.2) by x and substitute these identities. The two main terms t cancel against $2t$, leaving $tf(t) + (f*\nu)(t) = O(1)$ uniformly for $t \ge \log 2$. The bounded initial interval is harmless. □

10 The local-density hypothesis from a quadratic sieve

The sieve below is Selberg's quadratic sieve [18], in a coarse form. Sharp constants in the resulting Brun–Titchmarsh inequality are due to Montgomery and Vaughan [19]; we need only the order of magnitude.

Lemma 10.1 (A coarse uniform interval sieve)**.** *For $X \ge 1$ and $Y \ge 2$,*

$$\pi(X+Y) - \pi(X) \ll \frac{Y}{\log Y}. \tag{10.1}$$

Proof. For an integer $z \geq 1$, write $G_z = \sum_{\ell \leq z} \mu(\ell)^2/\varphi(\ell)$. Choose real weights λ_d supported on $d \leq z$, with $\lambda_1 = 1$. The identity $\gcd(d,e) = \sum_{\ell \mid d,\, \ell \mid e} \varphi(\ell)$ gives

$$Q(\lambda) := \sum_{d,e \leq z} \frac{\lambda_d \lambda_e}{\operatorname{lcm}(d,e)} = \sum_{\ell \leq z} \varphi(\ell) \left(\sum_{\substack{d \leq z \\ \ell \mid d}} \frac{\lambda_d}{d} \right)^2. \tag{10.2}$$

Put $x_\ell = \sum_{\ell \mid d,\, d \leq z} \lambda_d/d$. Inversion gives $\lambda_1 = \sum_{\ell \leq z} \mu(\ell) x_\ell$. Cauchy–Schwarz therefore proves $Q(\lambda) \geq 1/G_z$, with equality at $x_\ell = \mu(\ell)/(\varphi(\ell) G_z)$. Inverting these minimizing values,

$$\lambda_d = \frac{\mu(d) d}{\varphi(d) G_z} \sum_{\substack{k \leq z/d \\ (k,d)=1}} \frac{\mu(k)^2}{\varphi(k)}.$$

Since the inner sum is at most G_z, $|\lambda_d| \leq d/\varphi(d) \leq d$ and $\sum_{d \leq z} |\lambda_d| \leq z^2$.

For each prime $n > z$, $\sum_{d \mid n} \lambda_d = 1$. Summing the nonnegative square of this divisor sum on $(X, X+Y]$ gives

$$\pi(X+Y) - \pi(X) \leq z + \frac{Y}{G_z} + O(z^4). \tag{10.3}$$

The error comes from the uniform $O(1)$ error in counting multiples of $\operatorname{lcm}(d,e)$.

At least one quarter of the integers through any integer n are squarefree: the nonsquarefree count is at most $n \sum_{k \geq 2} k^{-2} < 3n/4$. Partial summation gives $G_z \geq \sum_{\ell \leq z} \mu(\ell)^2/\ell \gg \log z$ for large z. Take $z = \lfloor Y^{1/8} \rfloor$. The z and z^4 terms are absorbed by $CY/\log Y$ for large Y, and the bounded remaining range is absorbed by increasing C. This proves (10.1). □

Proposition 10.2. *The prime-power measure in* (1.1) *satisfies* (2.4).

Proof. For large t and $e^{-t/2} \leq h \leq 1$, put $X = e^t$, $Y = e^t(e^h - 1)$. Then $\log Y \geq t/2$ and $Y \ll h e^t$. Lemma 10.1 gives

$$\sum_{e^t < p \leq e^{t+h}} \frac{\log p}{p} \ll h.$$

Prime powers of exponent at least two contribute $O((1+t)^2 e^{-t/2})$: their count through e^{t+1} is $O((1+t)e^{(t+1)/2})$, and each weight in the interval is at most $(t+1)e^{-t}$.

For $0 < h < e^{-t/2}$, enlarge the interval to logarithmic width $e^{-t/2}$. Its prime contribution is $O(e^{-t/2})$, absorbed by the same error majorant. A larger constant covers the bounded initial range of t. Thus one may take $q(t) = C(1+t)^2 e^{-t/2}$. □

Propositions 9.3 and 10.2 discharge every arithmetic hypothesis of Theorem 2.1. The sieve is an upper-density estimate; it assumes no asymptotic prime count in short intervals.

11 The exact sparse Mertens certificate

Let $N \geq 12$ be an integer. Write $\mathcal{S}_N$ for the squarefree positive integers at most N, and

$$\mathcal{T}_N = \{n \in \mathcal{S}_N : n \text{ odd},\ N/2 < n \leq N\}, \qquad \mathcal{C}_N = \{n \in \mathcal{T}_N : n \text{ composite}\}.$$

The admissible labels are

$$\mathcal{B}_N = \{b : 5 \leq b \leq N/3,\ b \text{ squarefree},\ (b,6) = 1\}.$$

For any real coefficient vector $\alpha = (\alpha_b)_{b\in\mathcal{B}_N}$, set

$$(A\alpha)(n) = \sum_{\substack{b\in\mathcal{B}_N\\ b\mid n}} \alpha_b, \qquad n \in \mathcal{T}_N, \tag{11.1}$$

$$(D\alpha)(m) = \sum_{\substack{b\in\mathcal{B}_N\\ N/2<bm\le N\\ (b,m)=1}} \mu(b)\alpha_b, \qquad m \text{ odd and squarefree.} \tag{11.2}$$

The second vector vanishes for $m > N/5$. The matrices are sparse incidence matrices of the same multiplication pairs, with the label sign retained in the cofactor block.

Proposition 11.1 (Exact finite identity)**.** *For every such α,*

$$M(N) = \sum_{n\in\mathcal{T}_N} \mu(n)(1-(A\alpha)(n)) + \sum_{\substack{m\le N/5\\ m \text{ odd, squarefree}}} \mu(m)(D\alpha)(m). \tag{11.3}$$

Consequently

$$|M(N)| \le \mathfrak{B}_N(\alpha) := \pi(N) - \pi(N/2) + \|1-A\alpha\|_{1,\mathcal{C}_N} + \|D\alpha\|_1 . \tag{11.4}$$

Proof. Pair $n = m$ and $n = 2m$ for odd squarefree m with $2m \le N$. Their Möbius values cancel, so $M(N) = \sum_{n\in\mathcal{T}_N}\mu(n)$. Expanding $\sum_{n\in\mathcal{T}_N}\mu(n)(A\alpha)(n)$ and writing $n = bm$ gives the second sum in (11.3), since $\mu(bm) = \mu(b)\mu(m)$ on its support. Take absolute values. No admissible label divides a prime in $(N/2, N]$, so each such prime contributes exactly one to the first residual norm. This yields (11.4). □

The target estimate is

$$\|1-A\alpha\|_{1,\mathcal{C}_N} + \|D\alpha\|_1 \ll \frac{N}{\log N}. \tag{11.5}$$

No normalization or parity-balance constraint on α is required by the exact identity.

12 Explicit weights and the cofactor reduction

Put $L = \log N$. Retaining only admissible labels, choose

$$\alpha_p = \frac{(\log p)^2}{L^2}, \qquad \alpha_{pq} = \frac{2\log p\log q}{L^2} \quad (5 \le p < q), \qquad \alpha_b = 0 \text{ otherwise.} \tag{12.1}$$

These are the squarefree prime and semiprime terms in the classical second generalized von Mangoldt weights.

Lemma 12.1 (Coverage and coefficient mass)**.** *For the coefficients* (12.1)*,*

$$\|1-A\alpha\|_{1,\mathcal{C}_N} \ll N/L. \tag{12.2}$$

If $\rho_b = \#\{n\in\mathcal{S}_N : b \mid n\}/|\mathcal{S}_N|$, then $\sum_b \rho_b|\alpha_b| \le 1$.

Proof. For $n \in \mathcal{C}_N$, let $\epsilon_n = \mathbf{1}_{3|n}$. Expanding the square of the sum of logarithms of its prime factors at least five gives exactly

$$(A\alpha)(n) = \frac{(\log n - \epsilon_n \log 3)^2}{L^2} - \mathbf{1}_{\substack{n=pq \\ 3\nmid n}} \frac{2\log p \log q}{L^2}. \tag{12.3}$$

Every relevant prime or two-prime divisor is at most $N/3$ except the whole number $n = pq$ when $3 \nmid n$; this is the subtracted term. The first term has deficit at most $2\log 6/L$, since $N/2 < n \le N$. Also, by Chebyshev and partial summation,

$$\sum_{\substack{p<q \\ pq\le N}} 2\log p \log q \le 2 \sum_{p\le\sqrt{N}} \log p\, \vartheta(N/p) \ll N \sum_{p\le\sqrt{N}} \frac{\log p}{p} \ll NL.$$

This proves (12.2). For every squarefree $n \le N$, even or odd, the full retained divisor sum is nonnegative and at most $(\sum_{p|n,\ p\ge 5} \log p)^2/L^2 \le 1$. Averaging that inequality over $\mathcal{S}_N$ proves the mass bound. □

For an integer $Q \ge 1$ and real $x \ge 1$, define

$$G_Q(x) = -\sum_{\substack{p\le x \\ (p,Q)=1}} (\log p)^2 + 2 \sum_{\substack{p<q,\ pq\le x \\ (pq,Q)=1}} \log p \log q, \qquad G = G_1.$$

For odd squarefree $m \ge 3$, the cutoff $b \le N/3$ is automatic when $b \le N/m$. Hence

$$(D\alpha)(m) = \frac{G_{6m}(N/m) - G_{6m}(N/(2m))}{L^2}. \tag{12.4}$$

The row $m = 1$ is zero because all labels are at most $N/3$.

Lemma 12.2 (Coprimality removal)**.** *The aggregate error in replacing both occurrences of G_{6m} by G in* (12.4)*, before division by L^2, is $O(NL)$.*

Proof. The union bound and Chebyshev give

$$|G(x) - G_Q(x)| \le \sum_{\substack{p|Q \\ p\le x}} (\log p)^2 + 2 \sum_{p|Q} \log p\, \vartheta(x/p).$$

Enlarge the sum over m to all integers $3 \le m \le N/5$. For $p = 2, 3$ the total is $O(NL)$. The terms with $p \mid m$ in the second sum are bounded by

$$CN \sum_{m\le N/5} \frac{1}{m} \sum_{p|m} \frac{\log p}{p} \ll NL \sum_{p} \frac{\log p}{p^2} \ll NL.$$

For the first sum, $p \mid m$ and $p \le N/m$ imply $p^2 \le N$, and there are at most N/p^2 eligible m. Their total is $O(N \sum_p (\log p)^2/p^2) = O(N)$. Both series converge by comparison with sums over the integers. The same estimates apply at $N/(2m)$. □

Lemma 12.3 (The signed prime/semiprime sum)**.** *For $x \ge 2$,*

$$G(x) = -2x \log x\, f(\log x) + O(x). \tag{12.5}$$

Proof. Replacing prime sums by prime-power sums gives

$$G(x) = \sum_{n \le x} \big((\Lambda *_D \Lambda)(n) - \Lambda(n) \log n \big) + O(x).$$

For completeness, the convolution terms with at least one proper prime power cost at most $2 \sum_{p^a \le x,\ a \ge 2} \log p \, \psi(x/p^a) \ll x \sum_{p,a \ge 2} \log p / p^a = O(x)$. The prime diagonal and the proper-power terms in $\Lambda(n) \log n$ also cost $O(x)$; the crude bound $O(\sqrt{x} \log^3(2x)) = O(x)$ suffices. Selberg's formula and partial summation therefore give

$$G(x) = 2x \log x - 2\psi(x) \log x + O(x),$$

which is (12.5). □

13 Sampling, the tail estimate, and the arithmetic conclusions

Lemma 13.1 (Sampling with a logarithmic weight)**.** *For real $X \ge 4$, put $L_X = \log X$, $J_X = \lfloor X/2 \rfloor$, and*

$$\Phi_X = \sum_{m \le J_X} \frac{\log(X/m)}{m} \, |f(\log(X/m))|.$$

Then

$$\Phi_X \le \int_0^{L_X} t |f(t)| \, \mathrm{d}t + O(L_X). \tag{13.1}$$

Proof. Set $g_X(u) = \psi(X/u)/X - 1/u = f(\log(X/u))/u$. For $m \le u \le m+1$,

$$|g_X(m) - g_X(u)| \le \frac{\psi(X/m) - \psi(X/(m+1))}{X} + \frac{1}{m(m+1)}.$$

Multiply by $\log(X/m)$ and integrate over this unit interval. The first errors telescope at cost at most $L_X \psi(X)/X = O(L_X)$; the second errors also cost $O(L_X)$. Replacing $\log(X/m)$ under the integral by $\log(X/u)$ costs $O(1)$, since $|g_X(u)| \ll 1/u$ and $\log(u/m) \le 1/m$. The substitution $t = \log(X/u)$ gives an integral over a subinterval of $[0, L_X]$ because $J_X + 1 \le X$. Enlarging that interval proves the claim. □

The arithmetic instance of Theorem 2.1 and Cauchy–Schwarz give

$$\Phi_X \le \sqrt{L_X E(L_X)} + O(L_X) \ll L_X. \tag{13.2}$$

In particular there is no additional $\log \log X$ loss in this transfer.

Theorem 13.2 (Closure of the sparse certificate)**.** *The explicit coefficients* (12.1) *satisfy* (11.5) *uniformly for integers $N \ge 12$. Consequently $|M(N)| \ll N/\log N$.*

Proof. Lemma 12.3 gives, for $3 \le m \le N/5$,

$$|G(N/m) - G(N/(2m))| \le \frac{N}{m} \big(2 \log(N/m) |f(\log(N/m))| + \log(N/(2m)) |f(\log(N/(2m)))| + C \big).$$

Summing and using (13.2), once with $X = N$ and once with $X = N/2$, yields

$$\sum_{\substack{3\le m\le N/5\\ m \text{ odd, squarefree}}} |G(N/m) - G(N/(2m))| \ll N \log N. \tag{13.3}$$

The $O(N/m)$ errors sum to $O(N \log N)$ as well. Now (12.4) and Lemma 12.2, after division by L^2, give $\|D\alpha\|_1 \ll N/L$. Lemma 12.1 gives the other term in (11.5). Finally apply (11.4) and $\pi(N) \ll N/\log N$. □

Corollary 13.3 (A direct quantitative PNT)**.** *For $x \ge 2$, $\psi(x) = x + O(x/(\log x)^{1/3})$.*

Proof. The source equation and (2.4) imply $|f(t+h) - f(t)| \le C_0 h + q(t)$ for $0 < h \le 1$. Increase C_0 so that $|f(t)|/(4C_0) \le 1$. Write $A = |f(t)|$. If $A \ge 4q(t)$, then $|f(t+h)| \ge A/2$ for $0 \le h \le h_0 := A/(4C_0)$. Hence, for $t \ge 1$,

$$E(t+1) \ge \int_t^{t+h_0} u^2 |f(u)|^2 \,\mathrm{d}u \ge \frac{t^2 A^3}{16 C_0}.$$

The energy bound gives $A \ll t^{-1/3}$. If $A < 4q(t)$, the same conclusion follows from the exponential majorant of q. The range $\log 2 \le t < 1$ follows from boundedness of f. Putting $t = \log x$ proves the corollary, including jump locations under the right-continuous convention. □

Remark 13.4 (The certificate's floor). For every admissible coefficient vector, $\mathfrak{B}_N(\alpha) \ge \pi(N) - \pi(N/2)$. The PNT just proved, followed by partial summation, implies $\pi(N) - \pi(N/2) \sim N/(2\log N)$. Thus the absolute-value certificate itself cannot give $o(N/\log N)$, even with better control of its other terms. This is a limitation of the certificate, not a lower bound for $|M(N)|$. A stronger rate requires retaining further signed cancellation or changing the final certificate.

14 Finite eigenstructure, singular vectors, and scope

This section records the finite linear-algebra formulation. Its identities are exact, but the analytic theorem above should not be read as proving every natural finite inverse estimate.

14.1 Similarity to a diagonal operator

Allow a real endpoint $N \ge 2$ and put $J = \lfloor N/2 \rfloor$, $t_m = \log(N/m)$. On vectors indexed by $1 \le m \le J$, define

$$(S_N r)_m = t_m r_m + \sum_{\substack{k\ge 2\\ mk\le J}} \frac{\Lambda(k)}{k}\, r_{mk}, \tag{14.1}$$

$$(Z_N v)_m = \sum_{mk\le J} \frac{v_{mk}}{k}, \qquad (Z_N^{-1} v)_m = \sum_{mk\le J} \frac{\mu(k)}{k}\, v_{mk}, \qquad Q_N = \mathrm{diag}(t_m). \tag{14.2}$$

Finite divisor inversion gives the inverse in (14.2), and

$$S_N = Z_N^{-1} Q_N Z_N. \tag{14.3}$$

For instance the coefficient of v_{mk}/k in $Q_N Z_N - Z_N Q_N$ is $\log k$; the identity $\sum_{d\mid k} \Lambda(d) = \log k$ proves $Z_N S_N = Q_N Z_N$.

Thus the eigenvalues are the distinct positive numbers $t_m \geq \log 2$. Right eigenvectors are $Z_N^{-1} e_m$ and left eigenvectors are $e_m^{\mathsf{T}} Z_N$, with their natural biorthogonal normalization. This exact eigenstructure does not control the inverse norm in a scale-independent way: the similarity is not asserted to be uniformly well-conditioned.

For the actual vector $r_m = f(\log(N/m)) = \psi(N/m)/(N/m) - 1$, the elementary arithmetic inputs give

$$\|S_N r\|_\infty \ll 1. \tag{14.4}$$

This follows from (2.3) by omitting $k > N/(2m)$, where $f(\log(N/(mk))) = -1$. The omitted harmonic prime-power sum is $O(1)$ on a multiplicative interval of ratio two. The actual vector also satisfies

$$0 \leq 1 + r_m \leq C_1, \qquad \frac{1 + r_m}{m} \geq \frac{1 + r_{m+1}}{m+1} \quad (m < J), \tag{14.5}$$

because $(1 + r_m)/m = \psi(N/m)/N$.

14.2 The exact SVD expression and an obstruction

Let $\mathsf{W} = \mathrm{diag}(1/m)$ and

$$\mathsf{T}_N = \mathsf{W}^{1/2} S_N Q_N^{-1} \mathsf{W}^{-1/2}.$$

If $\mathsf{T}_N = U \Sigma V^*$ has singular values $s_j > 0$ and left singular vectors u_j, and $y = \mathsf{W}^{1/2} S_N r$, then

$$\sum_{m \leq J} \frac{t_m^2 |r_m|^2}{m} = \sum_{j=1}^{J} \frac{|\langle u_j, y \rangle|^2}{s_j^2}. \tag{14.6}$$

Indeed $\mathsf{T}_N(\mathsf{W}^{1/2} Q_N r) = y$. Equation (14.6) identifies the required interaction between small singular values and the distinguished source. It supplies no lower bound on every singular value.

There is a concrete obstruction to an unrestricted source estimate. Let $\lambda(n) = (-1)^{\Omega(n)}$ be the Liouville function and take $r_m = \lambda(m)$. Complete multiplicativity and Lemma 9.1 give, for $y \geq 1$,

$$\sum_{k \leq y} \frac{\Lambda(k)\lambda(k)}{k} = -\sum_{k \leq y} \frac{\Lambda(k)}{k} + 2 \sum_{p^{2a} \leq y} \frac{\log p}{p^{2a}} = -\log y + O(1).$$

Consequently $\|S_N r\|_\infty = O(1)$, whereas

$$\sum_{m \leq J} \frac{t_m |r_m|}{m} = \tfrac{1}{2} (\log N)^2 + O(\log N), \tag{14.7}$$

$$\sum_{m \leq J} \frac{t_m^2 |r_m|^2}{m} = \tfrac{1}{3} (\log N)^3 + O((\log N)^2).$$

Integral comparison proves the two sums. Thus bounded forcing alone cannot yield the desired source bounds. The Liouville vector meets the amplitude constraint but, for large N, violates the monotonicity in (14.5).

A sufficient spectral-tail estimate would be, for some fixed $\delta > 0$,

$$\sum_{s_j \leq s} |\langle u_j, y \rangle|^2 \leq C (\log N) s^{2+\delta}, \qquad 0 < s \leq 1. \tag{14.8}$$

Together with $\|y\|_2^2 = O(\log N)$, dyadic summation in (14.6) would give a discrete weighted energy of order $\log N$. The exponent 2 without a further summable factor does not suffice for that inference. Estimate (14.8) is a sufficient target, not a theorem established here.

14.3 The stronger constrained inverse remains open

The stronger statement would be that every real vector satisfying (14.5) and $\|S_N r\|_\infty \le C_0$ obeys

$$\sum_{m\le J} \frac{t_m|r_m|}{m} \le C(C_0,C_1)\log N. \tag{14.9}$$

Theorem 2.1 and the sampling argument prove the required bound for the actual arithmetic source. They do not prove (14.9) for all artificially constructed vectors: the analytic proof also uses the common positive measure and (2.4).

There is a finite, directly verifiable dual sufficient condition for (14.9). Define $(\mathcal{K}r)_m = r_m/m - r_{m+1}/(m+1)$ and $d_m = 1/[m(m+1)]$ for $m < J$. For every sign vector $\epsilon \in \{-1,1\}^J$ put $c_m = \epsilon_m t_m/m$. It would suffice to construct $a \in \mathbb{R}^J$ and nonnegative vectors u, v, z with $z \in \mathbb{R}^{J-1}$ such that

$$c = S_N^{\mathsf{T}} a + u - v - \mathcal{K}^{\mathsf{T}} z, \tag{14.10}$$
$$C_0\,\|a\|_1 + (C_1-1)\sum_m u_m + \sum_m v_m + \sum_m z_m d_m \le C\log N.$$

Pair the equality with r, use $\mathcal{K}r \ge -d$, $r \le C_1 - 1$, and $-r \le 1$, and then maximize over ϵ. This implication is just a finite inequality; no separation theorem is needed. A uniform construction of these multipliers has not been supplied.

15 Exact recursions and what they do not yet give

15.1 Restriction to divisibility rays

For an integer $d \ge 2$, put $X = N/d$ and $r_k^{(d)} = r_{dk}$ for $k \le \lfloor X/2 \rfloor$ (the identity below is trivial when $X < 4$). From (14.1),

$$(S_X r^{(d)})_k = (S_N r)_{dk}. \tag{15.1}$$

The amplitude and monotonicity constraints are preserved; the latter follows by restricting the nonincreasing sequence $(1 + r_m)/m$. This is exact recursive closure, but it provides no contraction factor.

15.2 Prime insertion in the coefficient family

For finitely supported coefficients on squarefree labels define

$$F_\alpha(X,m) = \sum_{\substack{b:\, bm\le X\\ (b,m)=1}} \mu(b)\alpha_b, \qquad H_\alpha(X,m) = F_\alpha(X,m) - F_\alpha(X,2m)$$

when the labels and m are odd. Let p be a fresh prime and split a new coefficient family into p-free branches $\alpha_0(b)$ at label b and $\alpha_1(b)$ at label pb. The child branch may contain $b = 1$. Direct divisor splitting gives

$$A_{\mathrm{new}}(n) = A_0(n) + \mathbf{1}_{p\mid n} A_1(n/p), \tag{15.2}$$
$$F_{\mathrm{new}}(X,m) = F_0(X,m) - \mathbf{1}_{p\nmid m} F_1(X/p,m). \tag{15.3}$$

For odd p, odd labels, and odd m,

$$H_{\mathrm{new}}(X,m) = H_0(X,m) - \mathbf{1}_{p\nmid m} H_1(X/p,m).$$

The minus sign is $\mu(pb) = -\mu(b)$; the exclusion $p \nmid m$ is necessary. The endpoint X/p is real. Replacing X by $\lfloor X \rfloor$ inside integer inequalities is harmless; dropping cofactor exclusions or renormalizing child masses is not.

15.3 Signed source folding

Let g be supported on squarefree integers coprime to a fixed excluded prime set, and let p be a new prime. For $p \nmid r$ define $(T_p g)(r) = g(r) - g(pr)$, with g zero beyond its support. Pairing r and pr gives

$$\sum_n \mu(n) g(n) = \sum_{p\nmid r} \mu(r)(T_p g)(r), \tag{15.4}$$

$$\|T_p g\|_1 = \|g\|_1 - 2 \sum_{\substack{p\nmid r \\ g(r)g(pr)>0}} \min(|g(r)|, |g(pr)|). \tag{15.5}$$

The function $T_p g$ remains supported on integers coprime to the excluded primes and to p. Equation (15.5) proves nonexpansion in ℓ^1. Strict contraction requires a lower bound on the displayed same-sign overlap. It does not follow from the singular values of a two-entry difference row.

The successful analytic argument does use a scale evolution $\sigma \mapsto \sigma/2$, but it has not established a fixed-dyadic contraction for these finite recursions. For a future rate improvement, the signed prime-boundary contribution must remain coupled to composite or cofactor sources before the final absolute values are taken. No such stronger recursive certificate is claimed here.

16 Relation to the literature

The literature comparison below was made in October 2026. Sources marked *read* were consulted in the original or in a scan. The arithmetic bounds are classical, and the pointwise form is Wirsing's (Section 16.2). What we compare is the method: Section 16.3 lists the known ways of getting a remainder from Selberg's formula and sets ours against them.

16.1 Selberg's formula and the elementary remainder problem

Selberg's symmetry formula [14] is the input to the Erdős–Selberg proofs of the prime number theorem. As reported in [12], Selberg remarked that his method, carried out carefully, also yields a remainder $\psi(x) = x + O(x/(\log x)^\alpha)$ for some $\alpha > 0$. The first explicit exponents, as listed in Wirsing's introduction [12], were $\alpha = 1/200$ (van der Corput), $\alpha = 1/10$ (Kuhn [10]), and $\alpha = 1/6$ (Breusch [6], by a method starting from a formula similar to Selberg's). Wirsing's first paper [12] gave $\alpha = 3/4$ (read). Bombieri [1] and Wirsing's second paper [13] obtained every α (both read). Later elementary work reached $x\exp(-c(\log x)^a)$: $a = 1/7 - \varepsilon$ by Diamond and Steinig, and $a = 1/6 - \varepsilon$ by Lavrik–Sobirov and Sokolovskiĭ; these are cited here from Diamond's survey [7] (read). Selberg himself noted [14, footnote 2] that his method also proves some of Beurling's theorems on generalized primes, with weaker results than Beurling's analytic ones.

Diamond [7, Section 8] explains the modest early exponents as a limitation of Selberg's formula itself. Every monotone function of the form $x/\log x + x/\log^2 x + O(x/\log^c x)$ with $c > 2$ satisfies the

generalized Selberg formula with remainder $O(x)$. Hence that formula cannot detect a ψ-error below the scale $x/\log x$. Bombieri overcame this with higher-weight analogues of the formula. Wirsing instead fed each new error estimate back into an improved Selberg formula.

16.2 Wirsing's estimate

Write $\rho(\xi) = \sum_{n \le e^\xi} (1 - \Lambda(n))/n - 2C$, with C Euler's constant. Wirsing [12, Sections 2 and 4] uses arithmetic convolution identities, harmonic and divisor-sum estimates, and positivity of Λ to supply the two relations

$$\xi\rho(\xi) = \int_0^\xi \rho(\xi - \eta)\,\mathrm{d}\rho(\eta) + O(1), \qquad |\rho(\xi) - \rho(\eta)| \le \xi - \eta + O(1/\eta) \quad (\eta \le \xi). \tag{16.1}$$

The first is a Selberg-type relation for this arithmetic integrated variable. The second is a two-sided fluctuation bound. He also notes that $\rho(\xi) = O(\xi^{-\alpha})$ gives $\psi(x) = x + O(x/\log^\alpha x)$ with the same α; this follows by partial summation.

In [12, Section 3] Wirsing states that (16.1) implies $\rho(\xi) = O(1/\xi)$, "das bestmögliche Ergebnis bei diesen Voraussetzungen" (the best possible result under these hypotheses). In arithmetic terms this is the pointwise bound

$$\psi(x) = x + O\left(\frac{x}{\log x}\right),$$

with the arithmetic input supplied in that paper and no sieve. It implies (1.2) and is strictly stronger. This does not by itself identify an equivalent input for an arbitrary pair (f, ν).

The distinction is visible in the input calculation [12, Section 4, pp. 207–208]. Writing $r(x) = \rho(\log x)$, Wirsing uses

$$s(x) := \sum_{m \le x} \frac{r(x/m)}{m} = \sum_{k \le x} \frac{\tau(k) - \log k - 2C}{k} = s(\infty) + O\left(\frac{\log x}{\sqrt{x}}\right).$$

Möbius inversion supplies a bounded remainder in the equation for r; subtracting the bounded harmonic convolution supplies the first relation in (16.1). These arithmetic identities are not among the base assumptions of Theorem 8.2.

In [13, Section 5] the statement is made abstract, as Satz 2. Let ρ be real-valued on $[0, \infty)$, not necessarily continuous, and assume, as $\xi \to \infty$,

- $\int_0^\xi |\,\mathrm{d}\rho| \ll \xi$;
- $|\rho(\xi) - \rho(\eta)| \le \gamma(\xi - \eta) + O(\eta^{-\beta})$ for $\eta < \xi$, with $\gamma < \sqrt{2}$;
- $\rho(\xi) = \xi^{-1} \int_0^\xi \rho(\xi - \eta)\,\mathrm{d}\rho(\eta) + O(\xi^{-\beta})$ with $\beta > -1$, except at the at most countably many ξ where the integral is undefined.

Then $\rho(\xi) \ll \xi^{-\beta}$. The example $\rho(\xi) = 2\sin\xi$ shows that $\gamma \le 2$ is not enough.

Wirsing gives the proofs of the abstract theorems only in outline ("Näher wollen wir auf die Beweise der beiden Sätze nicht eingehen"). For the arithmetic iteration he proves in full only the gain to some $\alpha' > \beta - 1$, which is all the iteration needs, and he indicates how to reach $\alpha = \beta$ by running the induction on the whole system. His arithmetic result $\psi(x) = x + O(x/\log^A x)$ for every A is proved in full. It already gives $E(T) = O(1)$.

A note added in proof to [13] replaces one of his lemmas by Selberg's sieve bound for primes in short intervals. So the use of a sieve upper density, as in Section 10, has a precedent; its role there differs from its role here (Section 16.3).

Remark 16.1 (Local density is not needed on the Satz 2 route). Lemma 8.1 derives $\nu((t,t+h]) \le 2h + O_A(1/t)$ for bounded h from the base hypotheses. This interval bound does not by itself bound $f * f$ in Wirsing's self-convolution equation. Theorem 8.2 supplies the additional estimate $f \in L^2(0,\infty)$, and Corollary 8.3 then applies Satz 2 to $g = -f$. This proves $E(T) = O(T)$ for each admissible pair without (2.4). The original proof with (2.4) remains a direct route with constants controlled by its hypothesis bounds.

Remark 16.2 (The primitive substitution does not give the bridge). In the arithmetic case Wirsing's variable is $t - B(t) + C_0$ up to an $O(e^{-t})$ term. For a general pair, the function $\rho(t) = t - B(t) + C_0$ need not satisfy his self-convolution equation, even under (2.4). For a fully specified example, put $\varepsilon = 1/4$ and

$$f(t) = -e^{-t} + \varepsilon \frac{\sin\log(1+t)}{1+t}, \qquad h = f + f', \qquad \nu = (1+h)\,\mathrm{d}t.$$

Here $|h| \le 3\varepsilon$, so the density is positive and bounded, and

$$F(t) = -1 + e^{-t} + \varepsilon(1 - \cos\log(1+t)), \qquad B(t) = t + \varepsilon\left(1 - \cos\log(1+t) + \frac{\sin\log(1+t)}{1+t}\right).$$

Both f and h are $O((1+t)^{-1})$. Hence $f * h = O(\log(2+t)/(1+t))$ and $R = tf + F + f * h = O(1)$. All the abstract hypotheses hold. But

$$\rho(t) = C_0 - \varepsilon + \varepsilon\cos\log(1+t) - \varepsilon\frac{\sin\log(1+t)}{1+t}$$

is bounded, has total variation $O(\log(2+t))$, and has distinct subsequential limits C_0 and $C_0 - 2\varepsilon$. Therefore $\rho * d\rho = O(\log(2+t))$, whereas $t\rho - \rho * d\rho = O(1)$ would force $\rho(t) \to 0$. No choice of C_0 gives that equation. This disproves the identification, not the pointwise conclusion for f or the replacement bridge in Section 8.

Remark 16.3 (Sharpness of the energy bound). Under the hypotheses of Theorem 2.1, $E(T) \asymp T$ can occur. Take, on the entire half-line,

$$f(t) = -e^{-t} + \frac{\sin t}{4(1+t)}, \qquad h = f + f', \qquad \nu = (1+h)\,\mathrm{d}t.$$

Then $f(0) = -1$, $|h| \le 3/4$, and the density is positive and bounded. The primitive F is bounded by Dirichlet's test, and the source equation gives $B = t + f + 1 + F = t + O(1)$. Since $f, h = O((1+t)^{-1})$,

$$|(f * h)(t)| \ll \int_0^t \frac{\mathrm{d}u}{(1+u)(1+t-u)} = \frac{2\log(1+t)}{t+2}.$$

Thus $R = tf + F + f * h = O(1)$ and (2.4) holds with $q = 0$. Finally,

$$E(T) = \frac{1}{16}\int_0^T \frac{t^2\sin^2 t}{(1+t)^2}\,\mathrm{d}t + O(1) = \frac{T}{32} + O(\log(2+T)).$$

This proves sharpness in the abstract class, including the stronger density hypothesis. It does not assert sharpness for the prime-power measure, which has additional arithmetic structure.

16.3 Mechanisms compared

The sources read use six distinct mechanisms to extract information from Selberg's relation, or to exclude zeros on the line.

(K) *Sup-norm iteration in the real variable.* This is the route of Selberg and Erdős, and of Kuhn [10] (read). An upper bound $|R(\eta)| \le K\eta$ on large intervals is improved by inserting it into the Selberg inequality. Each step loses a constant factor, which produces the small early exponents.

(I) *Isoperimetric estimates.* In Wirsing I [12] (read), and in Bang's proof described by Diamond, the Selberg integral is read as an area enclosed by the curve $(\sigma(t), \sigma(\xi - t))$, and an inequality between area and non-Euclidean perimeter bounds it.

(W) *Mean-square self-convolution.* Hilfssatz 1 of Wirsing II [13] (read) states the following. If $\limsup x^{-1} \int_0^x f^2 = F < \infty$, $\limsup x^{-1} \int_0^x g^2 = G < \infty$, and $h = x^{-1}(f * g)$ has vanishing Cesàro mean, then $\limsup x^{-1} \int_0^x h^2 \le FG/2$. The constant $\frac{1}{2}$ is sharp, with $\sin x$ extremal. Applied to repeated self-convolutions, this contracts normalized mean squares of derivatives of interpolants and drives Wirsing's iteration. This is not a bound on $\int_0^\infty |f|^2$ for our coupled source. Its application in Sections 3–4 of that paper already uses the self-convolution equation and fluctuation bound supplied by arithmetic identities. Bombieri's function-field note [5] (read) closes the qualitative argument the same way: with $A = \limsup m^{-1} \sum r_k^2$, the lemma gives $A \le A^2/2$, and $A \le 1$ then forces $A = 0$.

(B) *Higher-weight Selberg formulas with sign-change reduction.* Bombieri's memoir [1] (read) has three steps. First, Amitsur's symbolic calculus, with log acting as a derivation, generates classes of generalized Selberg formulas of arbitrary weight, together with a generalized "fundamental inequality". Second, Wright's substitution $x = e^\eta$ gives associated functions $V_k(\eta; f)$ with $V_k' = V_{k-1} - V_k$. Third, the main step (his Section VII) is a real-variable integral reduction after Selberg and Wright: when V_0 changes sign in an interval of length λ, its Lipschitz bound makes $\int |V_0|$ smaller than the majorant by a fixed factor $\chi < 1$. This is iterated through the higher-weight formulas, with Beta-function bookkeeping for the convolutions. The Laplace transform appears only in the proof of the convolution identity $V_k(f * g) = \int V_m V_n$ (his Theorem 5.3), through the convolution and uniqueness theorems on the real axis. The memoir contains no frequency analysis, no doubled-frequency positivity, and no flow in σ.

(Z) *Nonvanishing on the line.* The 3-4-1 inequality $3 + 4\cos\theta + \cos 2\theta = 2(1 + \cos\theta)^2 \ge 0$ shows that a zero at $1 + it_0$ would force a pole at $1 + 2it_0$; this uses analyticity of ζ there. In the argument after Wiener and Geller [11], presented by Diamond [7, Section 7] (read), a zero makes the twisted sum tend to -1, and a short-interval upper bound rules out the resulting concentration of primes near bad phases. These are qualitative boundary arguments. The 3-4-1 argument already couples a frequency to its double; that coupling alone is not a new device.

(T) *Asymptotic spectral characters.* Tao [15] (read in summary) excludes the parity obstruction by an $|\cos|$-averaging gap in an asymptotic Banach algebra. It is qualitative.

The mechanism of Sections 3–7 has the following components.

(M1) *Selberg's relation as a flow.* Along a horizontal ray $\sigma = ae^{-x}$, the window-normalized transform $\beta = \sigma b(\sigma + i\xi)$ satisfies the exact Riccati equation $\beta' = -\beta - \beta^2 + e$ (Section 3). Its defect

is $O(\sigma g(\xi))$. The algebraic identity behind it, $\Lambda \log + \Lambda *_D \Lambda = \mu *_D \log^2$ (equivalently $\zeta''/\zeta = (\zeta'/\zeta)' + (\zeta'/\zeta)^2$), is classical. Its use as a dynamical system in σ, with stable fixed point 0, unstable fixed point -1, and the frequency-dependent window $\sigma \le c/g(\xi)$ on which the defect is small, was not found in the sources read.

(M2) *Quantitative doubled-frequency exclusion.* (Z) closes the 3-4-1 argument with analyticity at $1 + 2it_0$. Section 3 closes it instead with the *same Riccati flow at frequency* 2ξ. If $\beta(\xi)$ stays near -1 for a time $\log 2$, the variance inequality forces $\operatorname{Re}\beta(2\xi) \ge 4/5$, and the flow at 2ξ then decreases $\operatorname{Re}\beta(2\xi)$ at a rate greater than 1, which is impossible. The result $|1 + \beta| \ge 1/100$ holds uniformly in ξ and in the window. No continuation and no information about a zeta function at $1 + 2it_0$ is used. The sources read have no analogue of this finite-time, uniform exclusion.

(M3) *Exact linearization.* The Möbius substitution $W = \beta/(1 + \beta)$ sends the unstable fixed point to infinity and turns the Riccati flow into $W' + W = e/(1 + \beta)^2$ exactly. This yields $\int_0^{a(\xi)} |b(u+i\xi)|\,\mathrm{d}u \le C$ uniformly in ξ, and hence a bounded variation-of-constants resolvent for the actual source (Section 4). The fractional-linear substitution itself is classical. Mechanism (W) contracts normalized mean squares of iterated convolutions. (M3) inverts the Selberg operator frequency by frequency, using the actual measure as its own multiplier. This combined quantitative use was not found in the inspected sources.

(M4) *Local density as a Fourier-tail and start-line budget.* In (Z), and in Wirsing's note added in proof, short-interval bounds limit concentration of primes. Here the bound (2.4) instead controls weighted translation moduli. Through Poisson averaging with frequency-dependent widths, it gives an L^2 budget on the line $\sigma = a(\xi)$ (Sections 5 and 6). Section 8 uses the weaker interval bound derived from positivity to obtain an unweighted L^2 estimate. This yields the energy conclusion through Wirsing's theorem. The direct weighted-tail proof still uses (2.4); the new route does not assert that the same weighted estimates hold with an arbitrary $O(1/t)$ density error.

(M5) *L^2 charging of the forcing.* The Selberg remainder R enters only through $\int |P|^2$, by Tonelli (Section 7); a pointwise charge would lose a logarithm. This is a standard device.

The proof contribution is the combination of (M1)–(M3) with the tail and source estimates: a quantitative, frequency-resolved, continuation-free treatment of Selberg's relation in the half-plane of absolute convergence. The comparison distinguishes this combination from the inspected real-variable and qualitative boundary arguments; it does not establish historical priority for the combination or its individual ingredients.

Scope of the search. The prime number theorem literature is too large to search exhaustively. The search covered Diamond's survey of elementary methods [7], which catalogs the Tauberian routes from Selberg's formula, together with the primary sources for each distinct remainder mechanism compared here (Kuhn, Wirsing I and II, Bombieri). We inspected all of Wirsing I, pp. 205–214, and Wirsing II, Sections 1–5, pp. 1–15, and found no matched supply of global L^2 control under precisely our base coupled-source assumptions. This is a bounded source finding, not a proof that no classical derivation exists. For the Bombieri memoir we read pages 1–6, 21–34, and 39–46 of the scan we used, not every page. The search did not cover every derivative paper. Ingham's review [3] was not consulted; Wright [4] and Amitsur [2] are known to us through Bombieri's use of them in (B). No conclusion about their unexamined contents is drawn.

16.4 Other related work

- Tao [15] (read in summary) uses Selberg's formula and positivity of the log-prime measure in an asymptotic Banach-algebra proof of the prime number theorem. It is qualitative and gives no rate.
- Saidak [16] (read) proves the reverse implication: a PNT remainder gives a sharpened Selberg formula. His Remark 2.5 notes that the elementary-remainder literature did not sharpen Selberg's lemma itself.
- Hildebrand [9] (first page read) proves $M(x) = o(x)$ without Selberg-type formulas, using a large-sieve inequality together with Mertens' estimates and a sieve upper bound for primes in short intervals.
- Debruyne and Vindas [17] develop Laplace-Tauberian remainder theorems for Beurling generalized primes.
- Kahane [20] proves an L^2 prime number theorem for Beurling systems, with an L^2 hypothesis on the integer-counting error and an analytic proof. It is not a statement about abstract Selberg-type relations.
- McNamara [21] gives a dynamical proof of the prime number theorem that also uses Selberg's formula, through an entropy-decrement argument over real averages. It uses neither the Laplace transform nor a flow in σ, and gives no rate.
- The monograph of Diamond and Zhang [8] treats Beurling systems and L^2 prime number theorems (Kahane), using both analytic and elementary methods. It is a compilation and was not consulted.

The author's earlier work. The certificate of Sections 11–13 continues the author's work on sparse matrix representations of the Mertens function [22, 23, 24, 25, 26]. The matrix of [22] has determinant $M(n)$, and its rows annihilate the Möbius vector. Section 7.1 of [26] notes that any vector z in the row space gives $M(n) = \mu^{\mathsf{T}}(\mathbf{1} - H^{\mathsf{T}}z)$, and names an explicit choice of coefficients with a uniform residual bound as the missing ingredient. Proposition 11.1 uses a different family of relations, $\mu(bm) = \mu(b)\mu(m)$ over divisor labels b, with an ℓ^1 residual, and Section 12 supplies explicit weights. The operators S_N, Z_N, and Q_N of Section 14 do not occur in those works.

16.5 Contribution

1. The method (M1)–(M3). Selberg's formula is treated as a Riccati equation along horizontal lines. Positivity at the doubled frequency keeps the solution away from its unstable fixed point, uniformly and in finite time. A Möbius substitution removes the quadratic term, leaving a linear equation with a small solution-dependent forcing, and gives a uniform resolvent. With (M4)–(M5), this is a complete proof of the energy bound under (2.4), without analytic continuation.
2. The L^2 estimate of Section 8, which needs no density hypothesis. It supplies the input that Wirsing's abstract theorem requires for a general positive measure. Wirsing obtained that input from arithmetic identities, which a general measure does not have (Remark 16.2).
3. The explicit passage from the energy bound to the sparse Mertens certificate (Sections 11–13).

We did not find (1) or (2) in the sources read. That search was limited, so priority is not settled.

17 Position of the result and remaining questions

The divisor identities, the prime and semiprime weights, Selberg's formula, and the quadratic sieve are classical. So are the arithmetic bounds, and the abstract theorem used in Corollary 8.3 is Wirsing's. This paper contributes the Riccati-flow method (M1)–(M3), the estimates it gives for the source, and the explicit chain to the sparse certificate.

Status:

- Proved here: the energy theorem, the density bound for primes, the cofactor transfer, and (1.3).
- Proved here without (2.4): $f \in L^2(0,\infty)$ (Theorem 8.2).
- Proved here, assuming Wirsing's Satz 2: the energy bound and $f(t) = O(1/t)$ without (2.4). The constant may depend on the particular pair (f, ν).
- Sharp: the energy bound cannot be improved in the abstract setting (Remark 16.3).
- Open: the general constrained inverse (14.9), the spectral tail (14.8), and a uniformly contracting finite recursion.
- Limit: the current certificate cannot go below the floor in Remark 13.4, so the method gives no better rate as it stands.

The next step is a signed certificate that can go below that floor, together with the estimate it would need. Larger numerical sweeps, or rewriting the same finite identity, will not supply that estimate.

A A circular source estimate and its transfer limitation

The following result records a direct singular-mode mechanism. It is not needed for the exponential-window proof, but explains why the source must stay linked to the measure.

Proposition A.1. *Let η be any probability measure on the circle $\mathbb{R}/T\mathbb{Z}$, $T > 0$. Let q be the mean-zero periodic solution of*

$$dq + q\,\mathrm{d}t = T\eta - \mathrm{d}t.$$

Then

$$\|q\|_2 \le \frac{15}{2}\,\|q + \eta * q\|_2\,. \tag{A.1}$$

Proof. Use normalized Fourier coefficients, and put

$$\omega_j = 2\pi j/T, \qquad z_j = \int e^{-i\omega_j t}\eta(\mathrm{d}t), \qquad w_j = (1+\omega_j^2)^{-1}.$$

For $j \neq 0$, $q_j = z_j/(1+i\omega_j)$; $q_0 = 0$. These coefficients define an L^2 solution since $\sum_j w_j < \infty$. Set $e_j = w_j|z_j|^2$ and $d_j = w_j|z_j(1+z_j)|^2$. Call j bad if $|1+z_j| < 1/6$. Good indices satisfy $e_j \le 36 d_j$. At a bad index, $\operatorname{Re} z_j < -5/6$, and positivity gives

$$\operatorname{Re} z_{2j} \ge 2(\operatorname{Re} z_j)^2 - 1 > 7/18 > 1/3.$$

Therefore $|z_{2j}| \geq 1/3$, $|1 + z_{2j}| \geq 4/3$, and $w_{2j} \geq w_j/4$, so $d_{2j} \geq (4/81)e_j$. The map $j \mapsto 2j$ is injective. For every integer $J \geq 1$,

$$\sum_{0<|j|\leq J} e_j \leq 36 \sum_{0<|j|\leq J} d_j + \frac{81}{4} \sum_{0<|j|\leq 2J} d_j \leq \frac{225}{4} \sum_{0<|j|\leq 2J} d_j.$$

This explicitly pays for any overlap between directly controlled and charged modes. Let $J \to \infty$ and apply Parseval to obtain (A.1). □

If η is a point mass at $T/2$, the operator $I + \eta*$ annihilates every odd Fourier mode. Thus (A.1) is not an all-vector singular-value bound. For the generated source, the even companion modes control the bad odd modes in aggregate. If $v = q + \varepsilon$ with $\|\varepsilon\|_2 \leq e_0$, then $\|I + \eta*\| \leq 2$ and (A.1) give

$$\|v\|_2 \leq \frac{15}{2} \|(I + \eta*)v\|_2 + 16e_0.$$

For the arithmetic measure, take large T away from its atoms and set

$$B_T = B(T), \quad \eta_T = \nu|_{[0,T]}/B_T, \quad c_T = T/B_T, \quad b_T = \frac{c_T(1 + f(T))}{1 - e^{-T}}.$$

On $0 \leq t < T$, define

$$q_T(t) = c_T f(t) + (c_T - 1) + b_T e^{-t}. \tag{A.2}$$

Its endpoint traces agree, and $dq_T + q_T\,\mathrm{d}t = T\eta_T - \mathrm{d}t$; integrating gives zero mean. Also $c_T = 1 + O(1/T)$ and $b_T = O(1)$, so

$$\int_0^T t^2 |q_T(t) - f(t)|^2\,\mathrm{d}t \ll T. \tag{A.3}$$

For the projection Π_J onto $0 < |j| \leq J$, the bound $|z_j| \leq 1$ gives

$$\int_0^T |q_T - \Pi_J q_T|^2\,\mathrm{d}t \leq \frac{T^3}{2\pi^2 J}, \qquad \int_0^T t^2 |q_T - \Pi_J q_T|^2\,\mathrm{d}t \leq \frac{T^5}{2\pi^2 J}.$$

These are analytic tail bounds, not finite-data extrapolations.

The circular residual still has a wrap term. Let $K_T = \eta_T*$, $e_T(t) = c_T - 1 + b_T e^{-t}$, and

$$W_T(t) = \int_{(t,T]} f(T + t - u)\nu(\mathrm{d}u).$$

Splitting the circular convolution gives, almost everywhere,

$$(I + K_T)q_T = c_T \left[\left(1 - \frac{t}{B_T}\right) f(t) + \frac{W_T(t) + R(t)}{B_T} \right] + e_T + K_T e_T. \tag{A.4}$$

Replacing this residual by R/B_T would omit both the early source and the wrap term. Proposition A.1 alone therefore does not close the weighted arithmetic target. Sections 3–7 give a complete alternative using exponential windows and retaining the initial-source term explicitly.

AI assistance

The arguments in this paper were produced with the assistance of AI models and checked by separate AI proof audits. Agreement between AI audits is not independent human verification, and the paper has not been peer reviewed.

References

[1] E. Bombieri, *Maggiorazione del resto nel "Primzahlsatz" col metodo di Erdős–Selberg*, Ist. Lombardo Accad. Sci. Lett. Rend. A **96** (1962), 343–350; and *Sulle formule di A. Selberg generalizzate per classi di funzioni aritmetiche e le applicazioni al problema del resto nel "Primzahlsatz"*, Riv. Mat. Univ. Parma (2) **3** (1962), 393–440.

[2] S. A. Amitsur, *Arithmetic linear transformations and abstract prime number theorems*, Canad. J. Math. **13** (1961), 83–109.

[3] A. E. Ingham, review of A. Selberg, *An elementary proof of the prime-number theorem*, and P. Erdős, *On a new method in elementary number theory which leads to an elementary proof of the prime number theorem*; reprinted in *Reviews in Number Theory as printed in Mathematical Reviews 1940–1972*, vol. 4, N20-3, 191–193, Amer. Math. Soc., 1974.

[4] E. M. Wright, *The elementary proof of the prime number theorem*, Proc. Roy. Soc. Edinburgh Sect. A **63** (1952), 257–267.

[5] E. Bombieri, *Sull'analogo della formula di Selberg nei corpi di funzioni*, Atti Accad. Naz. Lincei Rend. Cl. Sci. Fis. Mat. Natur. (8) **35** (1963), 252–257.

[6] R. Breusch, *An elementary proof of the prime number theorem with remainder term*, Pacific J. Math. **10** (1960), 487–497.

[7] H. G. Diamond, *Elementary methods in the study of the distribution of prime numbers*, Bull. Amer. Math. Soc. (N.S.) **7** (1982), 553–589.

[8] H. G. Diamond and W.-B. Zhang, *Beurling Generalized Numbers*, Math. Surveys Monogr. **213**, Amer. Math. Soc., 2016.

[9] A. Hildebrand, *The prime number theorem via the large sieve*, Mathematika **33** (1986), 23–30.

[10] P. Kuhn, *Eine Verbesserung des Restgliedes beim elementaren Beweis des Primzahlsatzes*, Math. Scand. **3** (1955), 75–89.

[11] N. Wiener and L. Geller, *Some prime-number consequences of the Ikehara theorem*, Acta Sci. Math. (Szeged) **12B** (1950), 25–28.

[12] E. Wirsing, *Elementare Beweise des Primzahlsatzes mit Restglied. I*, J. Reine Angew. Math. **211** (1962), 205–214.

[13] E. Wirsing, *Elementare Beweise des Primzahlsatzes mit Restglied. II*, J. Reine Angew. Math. **214/215** (1964), 1–18.

[14] A. Selberg, *An elementary proof of the prime-number theorem*, Annals of Mathematics (2) **50** (1949), 305–313.

[15] T. Tao, *A Banach algebra proof of the prime number theorem*, 25 October 2014. Online exposition.

[16] F. Saidak, *On the prime number lemma of Selberg*, Mathematica Scandinavica **103** (2008), 5–10. doi:10.7146/math.scand.a-15065. An author preprint is available at `https://uncg.edu/~f_saidak/MS2007.pdf`.

[17] G. Debruyne and J. Vindas, *On General Prime Number Theorems with Remainder*, arXiv:1601.05324, version 2, 2017. `https://arxiv.org/abs/1601.05324`. Published in *Generalized Functions and Fourier Analysis*, Oper. Theory Adv. Appl. **260**, Birkhäuser, 2017, 79–94, doi:10.1007/978-3-319-51911-1_6.

[18] A. Selberg, *On an elementary method in the theory of primes*, Norske Vid. Selsk. Forh. (Trondheim) **19** (1947), 64–67.

[19] H. L. Montgomery and R. C. Vaughan, *The large sieve*, Mathematika **20** (1973), 119–134, doi:10.1112/S0025579300004708.

[20] J.-P. Kahane, *Sur les nombres premiers généralisés de Beurling. Preuve d'une conjecture de Bateman et Diamond*, J. Théor. Nombres Bordeaux **9** (1997), 251–266, doi:10.5802/jtnb.201.

[21] R. McNamara, *A dynamical proof of the prime number theorem*, Hardy–Ramanujan J. **44** (2021), doi:10.46298/hrj.2022.8924.

[22] J. Kline, *A sparser matrix representation of the Mertens function*, Linear Algebra Appl. **581** (2019), 354–366, doi:10.1016/j.laa.2019.07.021.

[23] J. Kline, *On the eigenstructure of sparse matrices related to the prime number theorem*, Linear Algebra Appl. **584** (2020), 409–430, doi:10.1016/j.laa.2019.09.022.

[24] J. Kline, *Bordered Hermitian matrices and sums of the Möbius function*, Linear Algebra Appl. **588** (2020), 224–237, doi:10.1016/j.laa.2019.12.004.

[25] J. Kline, *The smallest singular value of a sparse Mertens matrix*, research release v0.1.0, 2026, doi:10.5281/zenodo.21774716.

[26] J. Kline, *The singular spectrum of a sparse Mertens matrix*, research release v0.1.0, 2026, doi:10.5281/zenodo.23004980.